Engineered features, not extra stages, drive RuseTect

RuseTect's measurable detection power comes from **31 past only account history and network features inside one XGBoost model**, not from the number of layers around it. The brief's starting assumption is wrong on one point: **the first stage (L1) is an XGBoost gatekeeper, and LightGBM appears only as one half of the L2 average and in the L0 onboarding model** (rusetect_engine.py). On IBM AMLworld HI-Small test weeks the live path scores **PR-AUC 0.711 and ROC-AUC 0.985**, but on LI-Small it falls to **0.345**, and the only truly out of distribution figure (models never trained on LI) is **0.336** (full_eval_HI.json, retrain_extended.json). The block band a bank would see today is **63.8% precise at 65.3% recall**, not the 91% to 97% figures the Round 1 submission quoted, which came from an offline stacker that is not served (live_path_audit.json). Every training source is synthetic, simulated, authored or foreign, and no public, labelled, real UPI transaction dataset exists, so the next gains come from cross generator testing (AMLworld Medium and Large, SAML-D, AMLgentex), real label sanity checks on non UPI graphs (Elliptic++, DGraph-Fin, Libra Bank), and institutional access. The literature supports adding subgraph pattern features to the trees, calibrating each model, and reporting alert budget and cost metrics; it does not support replacing the trees with a graph neural network. The project's strongest research asset is its discipline (chronological splits, train and serve parity, preregistered adoption rules, honest negative results), and the jury conversation should trade that discipline for data access and deployment constraints.

Phase 1: two learned stages, not seven, make the payment decision

What the code actually runs

The live payment path is a two stage cascade. **L1 is an `xgb.Booster` with 30 trees; if its score is below 0.0563 the payment is approved and exits. Otherwise L2 risk is the plain mean of an XGBoost and a LightGBM score** on the same 31 features, and the result is banded at verify 0.1366, hold 0.4495 and block 0.8230 (rusetect_engine.py lines 76 to 149, cascade_config.json). Features are computed from account state before that state is updated with the current payment, so they are past only with respect to labels, and the training features were built by streaming through the same serving state code in time order (RETRAINING.md). LightGBM is served from JSON by a pure NumPy evaluator (README.md). Reason chips are TreeSHAP contributions from **the L2 XGBoost booster only**, so they explain half of a score that is an average of two models (rusetect_engine.py lines 150 to 154). L4 is a deterministic union find "ring linker" over payments in the high band, not a learned model.

Around this core sit layers that are implemented and unit tested but mostly not measured on real signal: a message model (logistic regression on hashed TF-IDF with a six rule hard signal floor), lure to payment fusion, context rules, session coercion weights (screen share 0.30, paste 0.20, hesitation 0.20, SIM swap 0.15, app hidden 0.10, network 0.05, step up at 0.65), typing and mouse verifiers, campaigns, an Ed25519 signed, hash chained consortium ledger of HMAC tokens (not a blockchain), and a two analyst review desk with a template based STR draft (SESSION_AND_NETWORK.md, model-selection.md). The Random Forest, the L3 sequence blend, the GNN, the logistic stacker with isotonic calibration and the autoencoder novelty queue are **offline only**; the config still carries an unused `l3_sequence_weight: 0.1` (project-audit.md). The codebase is about 7,584 lines of Python across service, scale and ml, with 53 test functions in 8 files; the tests were not run in this audit, so pass status is unverified. Calling the system a "seven layer" or "multistage ML" cascade overstates it: only two learned stages decide a payment.

What has been validated, and on what

The payment protocol is cleaner than most competition work. Data is split 60/20/20 in time order, thresholds and model choices are fixed on validation before test is read, and model swaps follow adoption rules written in advance (for example “HI at least current minus 0.005 and LI at least current plus 0.01”) (RETRAINING.md). The table collects the numbers that matter. False positive counts and rates marked “derived” are computed from the reported flags, precision and recall, using 1,013,872 HI test negatives and 1,383,885 LI test negatives.

Measure (live served path)	HI-Small test (1,015,669 payments, 1,797 laundering, 0.177%)	LI-Small test (1,384,810 payments, 925 laundering)
PR-AUC / ROC-AUC	**0.7107 / 0.9852**	**0.345 / 0.9674**
Out of distribution (HI-only models on LI)	not applicable	0.3355 / 0.9603
L1 pass rate and laundering kept	20.32% passed to L2, 99.17% kept	not reported
Verify or above	15,319 flags (1.508%), recall 0.886, precision 0.104; about 13,727 FP, FPR about 1.35% (derived)	19,100 flags, recall 0.718, precision 0.035; about 18,436 FP, FPR about 1.33% (derived)
Hold or above	6,773 flags (0.667%), recall 0.811, precision 0.215; about 5,316 FP (derived)	recall 0.596, precision 0.072
Block	1,840 flags, TP 1,173, FP 667, FN 624, precision 0.638, recall 0.653, F1 0.645	1,146 flags, recall 0.36, precision 0.291
Block retuned on validation (0.9345, not applied)	1,059 flags, TP 993, FP 66, precision 0.938, recall 0.553, F1 0.695	not computed
Ring recall (at least one payment flagged)	191/192 verify, 179/192 block; BIPARTITE weakest at 16/21	60/60 verify, 54/60 block

Sources: full_eval_HI.json, full_eval_LI.json, live_path_audit.json, retrain_extended.json.

Model selection evidence is broad. A 14 model comparison on HI-Small ranks **XGBoost AP 0.728, LightGBM 0.725, Random Forest 0.706, CatBoost 0.705**, down to logistic regression 0.169 (model_comparison_tabular.json). Sequence models (TCN 0.673, LSTM 0.671, Transformer 0.666, 1D CNN 0.661) all trail the trees, and no blend passed the validation gain rule (sequence_blend_check.json). The causal GNN reached 0.516 and Isolation Forest 0.004 (model-selection.md). The feature ablation is the single most important result in the repo: **raw transaction fields 0.180, plus sender behaviour 0.540, plus network features 0.648, all features 0.728** (layered_engine.json). Outside payments, L0 onboarding on Feedzai BAF reaches test ROC-AUC 0.892 to 0.897 on Base with age FPR parity 0.88 to 0.93 (retrain_l0_variants.json); the message layer reaches precision 0.951 and recall 0.677 in its high band on Indian rows (message_model_eval.json); the Aalto typing verifier reaches ROC-AUC 0.890 on one sentence for 2,868 unseen people (aalto_eval.json). Latency was measured at **0.33 ms p50 model only** on earlier models, **API p95 71.8 ms at 220 requests per second** on one worker, **41,459 payments per second** single process with compact state, and **Redis scale out p95 794.6 ms at 789 per second** on two shared cores (online_parity.json, load_test.json, scale_bench.json, scale_out_bench.json).

What works, what is incomplete, and where it bottlenecks

What works end to end is the payment API with the two stage cascade, streaming feature state that is bit identical in compact form on all 5,078,345 HI payments, TreeSHAP reasons, the ring linker and the review desk. What is incomplete is listed by the project’s own audit: the **block threshold mismatch is open with “decision needed”**, because 0.8230 was carried over from the offline L3 blend and applied to a different live score (project-audit.md finding 1).

State lives in memory, so on Vercel a message and the next payment can land on different instances and “the fusion can silently do nothing” (finding 5). Lure to payment fusion is only unit tested because no linked message and payment pairs exist (datasets.md). There is no call audio or voice detection, session signals have no public benchmark, the docker compose path is untested, and SAML-D was proposed but never run (SCALING.md).

Several documents are stale after the LI-Small retrain, and a jury reading them literally would be misled. `full_eval_LI.json` still says LI was “never used for training” although its 0.345 comes from the HI plus LI retrained models; SCALING.md still quotes 0.716 and 0.336 for the previous models; model-selection.md says L4 “turns 1,105 alerts into 318 cases” while checks.json says 1,140 into 349; and the config’s validation traffic shares (0.13% / 0.31% / 0.50%) disagree with the audit’s 0.443% hold or block and 0.945% step up or above (full_eval_LI.json, checks.json).

The architectural bottlenecks are concrete. The live score is an **uncalibrated mean** of two boosters trained with 10% negative subsampling and `scale_pos_weight`, so thresholds do not transfer between models, which is exactly how the block mismatch arose. Reasons explain only the XGBoost half, and **LightGBM is now the weaker half** (HI test 0.604 alone versus 0.711 for XGBoost) (l2_halves_check.json). A missing `payment_format` defaults to “ACH” and a missing currency to US Dollar (server.py lines 131 to 140); since payment format is a known shortcut feature (removing it drops PR-AUC from 0.728 to 0.701), an incomplete record can be scored as higher risk for an artefactual reason (checks.json). Finally the distributed path is two orders of magnitude slower in tail latency than the single process path, so the Redis design is a floor, not a capacity claim.

What a banking or ML expert would attack first

An experienced reviewer would note that **no metric is confidence bounded**: there is no bootstrap, seed variance or interval anywhere, so differences such as 0.7107 versus 0.7108 or 0.725 versus 0.728 are inside unknown noise. They would note that the IBM HI-Small test covers **only 2022/09/01 to 2022/09/18 of simulated time**, so “chronological” means an 18 day horizon on one generator, and that prevalence shifts sharply inside it (1,082 laundering in validation versus 1,797 in test of equal length; validation AP 0.633 versus test 0.711) (live_path_audit.json). They would find that the rules challenger is degenerate, with thresholds like “sender made at least 20,550 payments in 24 h” and “amount at least $3,808,605”, two of which fire on 1 and 2 test payments, so any “15x fewer alerts than rules” claim is against a straw man (layered_engine.json). They would see there is **no cost sensitive evaluation, no amount weighted recall, no ring level precision and no payment side adversarial test**. And they would point out the context and graph checks “were designed after looking at the bundled sample”, so their small gains are optimistic (full_eval_HI.json caveats).

Where the genuine research contribution lies

The publishable value is in evaluation methodology and honest negative results rather than in a new model. The repo demonstrates **train and serve parity by construction** (features identical on 5,078,345 payments; Bloom index decisions agree 100%), a **strictly causal benchmark of 14 tabular and 4 sequence models** on AMLworld with validation only selection, **preregistered adoption rules** for every swap, an **unseen typology holdout** in which XGBoost keeps 0.80 to 0.85 recall on removed typologies while Isolation Forest recovers 0.0 on all four, a BAF fairness control (bias features x1 and x2 raise ROC-AUC to 0.954 but collapse age parity to 0.185), and a PaySim shortcut diagnosis (0.52 without balances, 0.9997 with payer balance) (scale_bench.json, baf_variant_eval.json, paysim_eval.json). The threshold transfer bug is itself a cautionary case study worth writing up.

Phases 2 and 3: no public labelled UPI data exists, so diversity must come from elsewhere

Real, anonymised, synthetic and simulated are four different grades of evidence

The search confirmed a hard fact the jury should hear early: **no public dataset of real UPI transactions with fraud labels exists**. The one Kaggle "UPI" set inspected in detail, UPI Transactions 2024 (250,000 rows, 480 frauds, 0.19%), is described by its own analysis repo as "a simulated (synthetic) dataset" with no confirmed payer or payee IDs (analysis repo). RBI treats MuleHunter.AI outcome data as fiduciary and refused under RTI to disclose even the count of flagged mule accounts (MediaNama). Even a 2026 study of mobile money laundering in Rwanda had to use synthetic SAML-D for lack of real data (arXiv 2608.15447).

The candidates therefore fall into four grades, and the report keeps them separate throughout. **Real, non anonymised public records** are the public blockchains: the Ethereum phishing network and the Elliptic family's underlying Bitcoin graph. **Real, anonymised records** are DGraph-Fin (Chinese consumer lending), Libra Internet Bank (Romania), Berka PKDD99 (Czech bank), T-Finance and T-Social, ULB credit card, IEEE-CIS and the Elliptic feature sets (obfuscated features over real transactions). **Synthetic data derived from real data** includes SynthAML (real Spar Nord alerts synthesised with SDV), the RBA login dataset (synthesised from real login behaviour, "totally artificial" values), and BankSim (calibrated on aggregate Spanish bank data). **Simulated environments** with generator ground truth are AMLworld, SAML-D, AMLgentex, AMLSim, PaySim, Sparkov, TabFormer, the JPMorgan synthetic sets, LDBC FinBench and the Kaggle UPI uploads. Only simulated sets give mule style payment graphs with labels; only real sets test whether methods survive noisy, human assigned labels.

Verified catalogue, part one: identity, scale and class

Figures are from the primary paper or official page unless marked "unverified". Kaggle pages could not be opened during the research, so Kaggle licences and file formats are the least verified fields.

Dataset	Evidence grade	Source and link	Country, domain	Volume	Fraud or illicit cases	Class
AMLworld HI/LI-Medium	Simulated	Altman et al., NeurIPS 2023 (arXiv); Kaggle ealtman2019	Virtual world, multi bank transfers	31M to 32M txns, about 2M accounts	35K (1 in 905) HI; 16K (1 in 1,948) LI	**A**
AMLworld HI/LI-Large	Simulated	as above	as above	176M to 180M txns	223K (1 in 807) HI; 100K (1 in 1,750) LI	**A** (compute heavy)
AMLgentex (Swish, Bizum, Venmo)	Simulated, calibrated to public aggregates	AI Sweden, Handelsbanken, Swedbank (arXiv); Hugging Face	Sweden, Spain, US; instant P2P	100K or 1M accounts, 12 institutions	laundering with injected label noise	**A/B**
SAML-D	Simulated	Oztas et al., IEEE ICEBE 2023 (paper)	UK focused, cross border	9,411,384 txns per paper (9,504,852 per arXiv 2608.15447); 749,507 accounts	11,658 (0.124%), 28 typologies (17 suspicious)	**A/B**
IBM AMLSim	Generator	AI4FCF list	Generic	sample 10,000 users, 1.32M txns	SAR patterns	**B**

Dataset	Evidence grade	Source and link	Country, domain	Volume	Fraud or illicit cases	Class
JPMorgan synthetic payments and customer journey	Simulated	JPMorgan AI Research	Generic; P2P with beneficiary, session events	configurable	configurable fraud probability	**B**
RBA login dataset	Synthetic from real logs	das-group; Zenodo	Country not stated; online service logins	33M logins, 3.3M users	account takeover label from incident team	**B**
Elliptic++ (actors and transactions)	Real, anonymised	git-disl	Bitcoin	822,942 wallets, 2.87M wallet edges; 203,769 txns	14,266 illicit wallets; 4,545 illicit txns	**C/D**
Elliptic2	Real, anonymised	Bellei et al.	Bitcoin	49.3M nodes, 196.2M edges	2,763 suspicious of 121,810 labelled subgraphs	**C/D**
DGraph-Fin	Real, anonymised	Finvolution, NeurIPS 2022 (arXiv)	China, consumer lending	3,700,550 nodes, 4,300,999 edges	15,509 defaulters (about 1.3% of labelled)	**C/D**
Ethereum phishing (XBlock)	Real, pseudonymous	Kanezashi et al.	Ethereum	2,973,489 nodes, 13,551,303 edges	1,165 phishing nodes	**C/D**
Libra Internet Bank graph	Real, anonymised	Graphomaly	Romania, retail bank	3 months aggregated; size not stated	internal alerts and reports	**C/D**
T-Finance	Real, anonymised	via GAD benchmark	Online transaction network	39,357 nodes, 21.2M edges	4.6% anomalies, expert labelled	**C**
SynthAML	Synthetic from real	Spar Nord, Scientific Data 2023 (PMC)	Denmark, bank AML alerts	about 20,000 alerts, over 16M txns	about 17% of alerts reported	**C**
LDBC FinBench	Simulated	VLDB 2025	Generic bank graph	up to about 4.4M accounts, 18.9M transfers (SF100)	none (isBlocked flags only)	**C** (systems)
CFDB (customer level)	Derived from simulated	arXiv 2404.14746	Derived from SAML-D, AMLworld	855,460 SAML-D customers	0.75% to 1.23% alerted	**C**
ULB credit card	Real, anonymised	Zenodo	Europe, cards	284,807 txns	492 (0.172%)	**C**
IEEE-CIS (Vesta)	Real, anonymised	via FDB	E-commerce card not present	561,013 train (FDB split)	3.5%	**D/C**
Sparkov (kartik2112)	Simulated	via FDB	US cards	1,296,675 train (FDB)	5.7% in FDB sample	**D/C**
IBM TabFormer	Simulated	Padhi et al.	US cards	24M txns	29,342 of about 2.4M samples	**D**
Fraud E-commerce	Real origin unverified	via FDB	E-commerce signups	120,889 train	10.6%	**C/D**
BankSim	Synthetic, calibrated	EMSS 2014; AI4FCF	Spain, card to merchant	594,643 txns	7,200 (1.21%)	**B/D** for P2M; **E** for mules

Dataset	Evidence grade	Source and link	Country, domain	Volume	Fraud or illicit cases	Class
Berka PKDD99	Real, anonymised	CTU repository	Czech bank, 1993 to 1998	1,090,086 rows in 8 tables	loan default only (76 bad of 682)	**D** (behaviour realism); **E** for fraud
Hinglish and English scam text (bolewara)	Partly real (UCI), partly handwritten	Hugging Face	India, SMS scams	3,787 messages	binary scam label	**A** (message layer)
BrainRun, HMOG, HuMIdb	Real behavioural	BrainRun; HMOG; HuMIdb	Mobile touch and sensors	2,218 users; 100 users; over 600 users	no fraud labels	**B/D**
UPI Transactions 2024 (Kaggle)	Simulated, generator unknown	analysis repo	India, labelled UPI	250,000	480 (0.19%)	**C** (demo only)
Other Kaggle “UPI” sets	Unknown	e.g. devildyno	India claimed	unverified	unverified	**C/E** until inspected
YelpChi, Amazon reviews	Real	GAD benchmark	Review spam	45,954 and 11,944 nodes	14.5%, 9.5%	**D/E** (architecture only)
Amaretto	Simulated injections	AI4FCF	Securities trading	29.7M txns	injected anomalies	**E** (wrong domain)

Verified catalogue, part two: fields, time, graph, licence and integration

Dataset	Features	Time	Graph	Format	Licence	Main limitation	RuseTect integration
AMLworld Medium/Large	Timestamp, source and target account and bank, amount, currency, payment type (arXiv)	16 days Medium, **97 days Large**	Full payer to payee	same schema as Small sets already ingested	CDLA-Sharing-1.0 (IBM/AML-Data)	Same generator as Small, so no new real signal	Drop in loaders; Large enables weekly walk forward and long history windows
AMLgentex	Placement, layering, integration; income and spending; label noise; per bank local view	112 daily steps, typology switch at step 56	Full, partitioned across 12 banks	not confirmed	not stated	Laundering, not explicit mule labels	Closest public analogue to UPI partial observability; drift and federated tests
SAML-D	Time, date, sender, receiver, amount, payment type, bank locations, currencies, label, typology (paper)	about 10 months simulated	Payer to payee	not confirmed	CC BY-NC-SA reported, unverified	UK payment types; no device or SIM	Second independent generator; per typology recall
RBA logins	IP, ASN, geography, user agent, device type, RTT, success, attack IP, ATO	Feb 2020 to Feb 2021	None	CSV per repo, not confirmed	CC BY 4.0	Values artificial, not linked to payments	Train a login and device risk prior for the session layer

Dataset	Features	Time	Graph	Format	Licence	Main limitation	RuseTect integration
JPMorgan payments	Sender and beneficiary IDs, accounts, countries, sector, amount, type, label	not shown	Payer to beneficiary	on request	not stated	Licence unclear	P2P beneficiary features; customer journey sessions
Elliptic++	183 txn features, 56 wallet features (opaque)	49 time steps	Wallet and transaction graphs	via Google Drive	not stated	Crypto; opaque features	Real label check for ring features and any GNN arm
Elliptic2	43 node, 95 edge features, mostly binned	not confirmed	Subgraph labels	via elliptic.co	not stated for data	Very large, crypto	Test subgraph (“shape”) detection akin to ring linking
DGraph-Fin	17 node features, 49.9% missing	relative edge times only; random split	Emergency contact edges	loadable via PyG	non commercial, registration	Default, not mules; social edges	Robustness of graph methods to missing data
Ethereum phishing	Edge timestamp and amount	Full	Full	Kaggle	unverified	Crypto	Chronological 80/20 edge split benchmark
Libra Bank	Cumulative amount and count per pair	3 months, aggregated	Customer to customer	zip download	not stated	No per transaction time	Rare real bank graph with alert labels
SynthAML	Alert ID, timestamp, direction, type, log size in DKK	quarter accurate	None	Figshare	article CC BY 4.0	No counterparties	Alert triage head (reported vs dismissed)
FinBench	Persons, accounts, media, transfers, loans	3 years	Rich	Spark generator	paper CC BY-NC-ND	No fraud labels	Graph store and ring query latency tests
ULB, IEEE-CIS, Sparkov, TabFormer	PCA or card fields; device for IEEE-CIS	yes (IEEE-CIS relative)	No payee side	Kaggle or GitHub	ULB mirror CC BY 4.0; others unverified	Card domain	Sequence pretraining (TabFormer, Sparkov), device features (IEEE-CIS)
bolewara Hinglish scam text	Message text, binary label	none	none	Hugging Face	CC BY 4.0	Small, cleaner than real SMS, no digital arrest	Direct addition to L-1 training and evaluation
Kaggle UPI 2024	Type, category, device, network, bank, state, amount, hour	yes	IDs not confirmed	Kaggle	none stated	Generator unknown	Demo only; check for single column AUC near 1.0

Which additions give genuinely new information

Ranked by new information per hour of work, the order is clear. **AMLworld HI-Medium** is the cheapest step (same loader, 32M rows fit a workstation) and gives a three day test window; **HI-Large** is the only AMLworld option with enough time span (97 days) for monthly walk forward validation (Altman et al.).

SAML-D is the cheapest independent generator: training on IBM and testing on SAML-D (or the reverse) directly measures how much of RuseTect's 0.711 is generator artefact. **AMLgentex** is the strongest new synthetic source because Swish and Bizum are instant P2P mobile rails and each of its 12 banks sees only its local view, which mirrors how a UPI payer PSP and payee bank each see half of a mule chain (Östman et al.). **RBA** and the **JPMorgan customer journey** set are the only public sources found for pre payment session behaviour, which matters because coerced, authorised scam payments often look normal at the transaction level. For real labels, Elliptic++ actors, DGraph-Fin, Libra Bank and Ethereum phishing should be used to check that ring features and any GNN arm survive noisy human labels, not as production training data. Card benchmarks largely duplicate BAF and PaySim style supervision; keep at most TabFormer or Sparkov for sequence pretraining and IEEE-CIS for device features. **No public dataset labels "mule accounts" explicitly**; mule labels must be derived (for example accounts on laundering paths with fan in then rapid fan out), following the CFDB customer aggregation method (arXiv 2404.14746), whose per customer counts for HI and LI-Small appear swapped relative to IBM's table and should be recomputed.

For UPI realism, the public anchors are aggregates. UPI handled **25.47 billion transactions worth Rs 31.53 lakh crore in October 2026, about 822 million per day** (Free Press Journal). I4C/NCRP data, via secondary reporting, puts 2025 citizen losses at **Rs 19,812.96 crore across 21,77,524 complaints, with investment scams 77% of losses and digital arrest 8%** (FastBull citing I4C). RBI's own fraud tables cover only frauds of Rs 1 lakh and above and show an implausible FY26 card and digital figure (293 cases, Rs 29 crore, against 13,332 cases in FY25), which looks like a reporting or classification change, so they should not size UPI scams (Outlook Business / PTI). A rough complaint rate of about 1 in 100,000 UPI transactions follows from these aggregates; this is an estimate, since complaints are not transactions. A UPI style simulator recalibrated on these aggregates (AMLSim or AMLgentex with single currency, account to account instant transfers, VPA and device fields) is the realistic route to "UPI data" without a partner.

Phase 4: graph features in trees beat graph networks under causal evaluation

The literature lines up with RuseTect's own negative results. On AMLworld, **FraudGT reports minority class F1 of 76.13 on HI-Small against 63.23 for XGBoost with graph features and 68.19 for Multi-PNA**, but the GNN advantage is not uniform (Multi-PNA collapses to 6.59 on Large-LI against 24.23 for XGBoost with graph features), the metric is F1 at a tuned threshold rather than PR-AUC, and message passing inside an evaluation window is not strictly past only per decision (FraudGT). When future graph context is removed, the picture flips: an April 2026 strict inductive reevaluation on Elliptic found **Random Forest F1 0.821 and XGBoost 0.775 beat GraphSAGE 0.688, GAT 0.610 and GCN 0.503**, transductive versus inductive GraphSAGE differed by 39.5 F1 points, and shuffled edges scored higher than the real graph (Maganti, arXiv 2604.19514). RuseTect's causal GNN at 0.516 against XGBoost at 0.72 fits that pattern. The best supported graph upgrade is the **Graph Feature Preprocessor**: past only fan in, fan out, scatter gather, gather scatter, simple and temporal cycle counts and windowed vertex statistics fed to boosted trees, reporting up to 36% minority class F1 improvement over GNN baselines and higher throughput on a 32 core CPU than GNNs on a V100 GPU (Blanuša et al., ICAIF 2024).

Technique	Problem it would solve in RuseTect	Evidence	Verdict
GFP style subgraph pattern features (past only)	Network features already add 0.108 PR-AUC (0.540 to 0.648); explicit mule shapes may add more and help BIPARTITE (16/21 at block)	GFP; FlowScope	**Add now**
Per model calibration before averaging (isotonic or Venn-Abers on a later holdout)	Uncalibrated mean caused the block threshold bug; boosters are poorly calibrated, more so after undersampling	Niculescu-Mizil and Caruana; Dal Pozzolo 2015; Venn-Abers	**Add now**

Technique	Problem it would solve in RuseTect	Evidence	Verdict
Alert budget and cost metrics (precision at k per day, recall at fixed FPR, amount weighted)	PR-AUC alone hides what an ops team sees; no cost evaluation exists	Saito and Rehmsmeier; Elkan; Bahnsen et al.; Fraud handbook	**Add now**
Time sliced evaluation with label delay; scheduled retraining	Production labels arrive late; 18 day test hides drift	Dal Pozzolo et al. TNNLS 2018; Lebichot et al.	**Add now**
TreeSHAP on the full served score plus rule and graph evidence	Reasons currently cover only the XGBoost half	Lundberg et al.; caveat Weerts et al., Jesus et al.	**Add now**, framed as investigator support
Sequence model embedding as one fused feature	Sequence models alone trail trees (0.66 to 0.67)	Nubank joint fusion +1.25% relative AUC, alone +0.08% (nuFormer); Stripe and Visa gains on billions of transactions (TechCrunch, TransactionGPT)	**Experiment**, low priority
Causal temporal GNN (TGN) or FraudGT research arm	Tests whether learned graph structure adds anything over GFP	TGN AUC 0.77 vs GraphSAGE 0.63 on DGraph with no tree baseline (arXiv 2404.00060); EdgeBank warning (Poursafaei et al.)	**Experiment** only against GFP plus XGBoost on identical splits
Simulated two bank partition	Quantifies ring recall lost when each bank sees half the chain	PETs Prize financial crime track (OSTP); AMLgentex local views	**Experiment**
Learning to defer with analyst capacity	Turns bands into a capacity aware review queue	Alves et al.	**Experiment** at Level 3
Unsupervised anomaly detection as a scorer	Was meant to catch novel patterns	ADBench: semi supervised with about 1% labels beats unsupervised (Han et al.); repo IF 0.004 and 0.0 on held out typologies	**Do not add**; keep only as a drift or novelty monitor
SMOTE style resampling	Imbalance	Resampling did not improve AUC and miscalibrated (van den Goorbergh et al.)	**Do not add**
Replace trees with GNN	Ring detection	Strict inductive evidence against (Maganti); Grinsztajn et al.	**Do not add**
Dedicated adversarial training module	Evasion	Thin, feature perturbation only (Cartella et al.; Lunghi et al.)	**Do not add**; run scripted evasion tests instead

Mapped onto the brief's eight questions, the evidence gives specific answers. Graph based account relationship analysis is worth it **as features**, not as a model. Temporal behaviour modelling is already the largest single gain (sender behaviour lifted PR-AUC from 0.180 to 0.540), so its next step is longer windows on HI-Large, not a new network. Velocity and sequence analysis should enter as GFP windowed statistics and at most one fused embedding. Unsupervised anomaly detection should be demoted to monitoring, since the autoencoder added only 25 catches for 507 alerts and Isolation Forest added none (postanalysis.json). Adaptive detection means rolling retraining with label delay and drift reporting rather than online learning. Explainable decisions need SHAP over the full score. Risk based prioritisation needs alert budgets and a capacity aware review band. Multi account correlation should move from "shared account in blocked payments" to "shared flow pattern" via scatter gather and short cycle counts around flagged accounts, with ring level precision reported.

Phase 5: the block band a bank would see is 64% precise, not 97%

Fifteen jury questions answered with the repo’s own numbers

Jury question	Honest answer	Repo evidence
1. Can it detect unseen patterns?	Partly, within one generator. Removing a typology from training leaves XGBoost recall of 0.800 to 0.853 at a 0.2% budget; the fused anomaly system does worse (0.716 to 0.830); Isolation Forest gets 0.0. Across generators it largely fails: HI-only models fall from 0.716 to 0.336 on LI, and the HI tuned ctx-1.3 mule rule catches 34 of 1,797 on HI but **0 of 925 on LI**	layered_engine.json; retrain_extended.json; context_tuning_LI_fixed.json
2. Does validation reflect banking conditions?	The protocol does (chronological, past only, validation only selection); the data does not. The horizon is 18 simulated days, labels are instant and perfect, there is no label delay and no seasonality	full_eval_HI.json
3. Is there leakage?	Label leakage is controlled by construction. Known leaks were measured and removed (PaySim balances 0.52 vs 0.9997; BAF x1, x2). Residual risks: payment format shortcut (0.728 to 0.701), context checks designed after seeing the sample, and the 24 h windows include the current payment’s own amount (not its label)	paysim_eval.json; checks.json
4. Severe imbalance?	Handled sensibly: all positives plus 10% of negatives, `scale_pos_weight`, PR-AUC and operating point reporting at 0.177% prevalence. The side effect is miscalibrated scores	RETRAINING.md
5. Precision and recall good enough?	Not yet. Verify or above is 89.6% false (precision 0.104); block is 0.638 precise with 624 missed laundering payments of 1,797. On LI, verify precision is 0.035, about 28 false flags per catch (derived)	full_eval_HI.json; full_eval_LI.json
6. False positive rate?	Derived from the reported counts: about 1.35% at verify, 0.52% at hold, 0.066% at block on HI; retuned block 0.0065%	as above, live_path_audit.json
7. Behaviour change over time?	Visible even inside 18 days: L1 pass rate 14.1% on validation vs 20.3% on test, laundering count 1,082 vs 1,797. BAF Base FPR drifts from 5.0% to 5.4% to 5.8% across months. No retraining schedule or drift monitor exists	live_path_audit.json; retrain_l0_variants.json
8. Coordinated multi account fraud?	Ring recall is high (191/192) but defined generously as one flagged payment per ring at a 1.5% flag rate. Ring level precision is not computed. The authored mule ring fixture stops only 9 of 24 and holds 0. The coordination rule adds 693 flags for 1 extra catch, so it is off	scenario_suite.json; coordination_eval.json
9. Legitimate unusual vs fraud?	Weakly. The context layer raises 963 payments to catch 1; IT and PF refund notices are flagged high 27.7% of the time in the message layer; the 1 in 200 genuine false alarm comes from authored scenarios designed together with the rules	full_eval_HI.json; message_model_eval.json

Jury question	Honest answer	Repo evidence
10. Is the multistage design justified?	Only L1 for compute: it removes 80% of L2 calls at a cost of 0.83% of laundering. Everything else is not justified on payment data	full_eval_HI.json
11. Does every stage add value?	No. LightGBM averaging: 0.7107 vs XGBoost alone 0.7108. Stacker: 0.7254 vs flat XGBoost 0.7279. Rules in fusion: +0.0005. Anomaly layer lowers unseen typology recall. Context layer 1 catch per 963. The value is in the feature families	l2_halves_check.json; layered_engine.json
12. Real time latency?	Model only 0.33 ms p50 (old models, 60,000 slice); API p95 71.8 ms at 220 rps; single process 41,459 per second; Redis distributed p95 794.6 ms at 789 per second. The retrained models' engine latency is not remeasured	load_test.json; scale_out_bench.json
13. Missing fields?	Silently imputed: format to ACH, currency to USD at rate 1.0. Since format is a shortcut feature this can shift risk for artefactual reasons; no missing field robustness test exists	server.py; rusetect_engine.py
14. Adversarial manipulation?	Not tested on payments (no structuring, delayed forwarding or perturbation test). Message paraphrase probes escalated only 2 of 5 scams and wrongly escalated 1 of 4 legit, and were not frozen before probing	challenge_results.json
15. Can an investigator understand a flag?	Partly. Top 4 TreeSHAP reasons are shown, but from the XGBoost half only, and the 0 to 1000 score is "a ranking aid, not a probability". Evidence that SHAP improves analyst accuracy is weak	SESSION_AND_NETWORK.md; Weerts et al.

What the metrics mean at UPI scale

The headline ROC-AUC of 0.985 is the least informative number in the repo, because at 0.177% prevalence a model can have near perfect ROC-AUC while nine of ten alerts are false. PR-AUC and operating points are the right lens, and they expose two problems. First, **calibration**: the only calibrated scores in the repo (Brier 0.00076, ECE 0.00025) belong to the offline stacker, and the live score has no calibration at all (layered_engine.json). Second, **prevalence**: HI-Small has one laundering payment per 565, while the I4C aggregates suggest roughly one complaint per 100,000 UPI transactions. If the block band kept the same detection and false positive rates at that prevalence, its precision would fall from 0.638 to roughly 1%, and even the retuned threshold would reach only about 8% (derived, rough, assuming unchanged rates). At 822 million UPI payments per day, a 1.5% verify rate would mean about 11 million step ups per day, and the average load alone is about 9,500 payments per second, ten times the Redis scale out measurement. These derived figures are illustrations, but they are the arithmetic a banking juror will do in their head, and RuseTect should do it first.

Per category performance is reported only by AMLworld typology (BIPARTITE weakest at 0.762 ring recall at block) and by message workflow (family emergency 0.50, parcel customs 0.582). No amount weighted recall exists, so it is unknown whether high value laundering is caught preferentially, and no cost model sets the price of a missed mule credit against a false hold on a genuine customer.

Random versus chronological validation

No random split comparison exists: searches for `train_test_split`, `StratifiedKFold` and `shuffle=True` in ml/ return nothing, and every payment result is chronological. That is a strength, not a gap in rigour, but it leaves the most persuasive slide unmade. Running the same pipeline with a shuffled split would quantify how much a typical random split benchmark overstates performance on AMLworld, and the Elliptic reevaluation's 39.5 point transductive versus inductive gap suggests the effect can be large (Maganti). It is a one day experiment and should be in the deck.

Claims in the Round 1 submission and research paper that the repo does not support

Claim	Where	What the repo shows
"97% of payments scored 0.9 or above were truly laundering (950 payments)" and "1,140 holds at 91% precision"	Round 1 PDF	Offline isotonic stacker only. The live block band is **0.638 precise on 1,840 alerts** (live_path_audit.json)
Decision bands on "calibrated probability" 0.10 / 0.40 / 0.90, and L3 labelled "LIVE: XGB + LGBM"	Round 1 slide 4	Served bands are **uncalibrated thresholds 0.137 / 0.450 / 0.823** (cascade_config.json)
"15x fewer alerts than a tuned rules engine (2,282 vs 34,128)"	Round 1 PDF	Rules baseline thresholds are degenerate; against one XGBoost the stacker needs 2,282 vs 2,375, only 3.9% fewer (checks.json)
"80% to 85% of a fraud pattern never seen in training"	Round 1 PDF	That is the XGBoost only range; the fused system scored 0.716 to 0.830 (layered_engine.json)
"0.33 ms median"	Round 1 PDF	60,000 payment slice with earlier models; API p95 is 71.8 ms (online_parity.json)
"L4 novelty watch"	Round 1 PDF	Autoencoder is offline; L4 is now the ring linker (README.md)
Ensemble "XGBoost 0.4 / Isolation Forest 0.2 / GNN 0.4" with GraphSAGE, on Kafka and Spark Structured Streaming	Research paper	Isolation Forest measured at 0.004 and dropped; GNN 0.516 offline only; streaming is Redis Streams (model-selection.md, SCALING.md)
"The graph channel is what converts five individually ambiguous signals into one high confidence detection"	Research paper case study	No GNN result supports this; hand built graph features inside XGBoost carry the signal (0.540 to 0.648)
Data schema with UPI channel, P2P/P2M, KYC tier, account opening date	Research paper	None of these fields exist in any dataset used (datasets.md)
Section 5.4 metrics "tracked rather than final benchmark numbers"	Research paper	The repo now has final numbers; the paper should report them
L0 "ROC AUC 0.892, FPR ratio 0.31 to 0.88"	Round 1 PDF	Superseded by better figures (0.897, parity 0.93); an improvement, not a contradiction
Prototype "still auto blocks the top band"	Round 1 slide 10	Fixed: blocks now need two analyst approval

The Round 1 brief explicitly weighs "awareness of false positives, bias, misuse, and adversarial adaptation" and says to "avoid automatic blocking or accusations based on a single weak signal". The paper should be rewritten around what the repo measured before any jury reads it, because it currently contradicts the project's own negative results.

Phase 6: methodology papers are realistic, a UPI fraud intelligence platform is ambitious

Direction	Scientific justification and research	Datasets	Technical requirements	Complexity	Expected benefit	Evaluation	Publication angle
1. Transaction intelligence (sequences)	Behaviour features already give the largest gain (0.180 to 0.540); fused embeddings add about 1% relative AUC at Nubank scale (nuFormer)	AMLworld Large (97 days), TabFormer, Sparkov for pretraining	Longer past only windows; optional TCN embedding as a feature	Low to medium	Better behavioural deviation; modest PR-AUC gain	Same chronological split, gain rule fixed in advance, bootstrap interval	Negative or small results on synthetic sequences are publishable with the causal protocol
2. Fraud network intelligence	GFP features beat GNN baselines with higher CPU throughput (GFP); FlowScope for multi step flows (AAAI)	AMLworld S/M/L, SAML-D, AMLgentex; Elliptic++ and Libra for real labels	Incremental cycle and scatter gather counters in streaming state	Medium	Higher ring recall at block, ring level precision, better BIPARTITE	Per typology recall, ring precision, latency per payment	“Strictly causal graph features vs GNNs on AMLworld” fills a gap no peer reviewed paper covers
3. Adaptive detection	Delayed labels and alert feedback matter in production (Dal Pozzolo 2018)	AMLworld Large walk forward; AMLgentex step 56 typology switch; BAF months	Retraining scheduler, label delay simulator, drift monitors	Medium	Measured decay curve and retraining cadence	Weekly walk forward with N day label delay; time to recover after drift	Drift and label delay benchmark on AML generators
4. Explainable financial crime detection	Regulators expect explainability (RBI FREE-AI via Mondaq); SHAP gains for analysts are not proven (Jesus et al.)	Existing results plus ring cases	SHAP on full score, graph path evidence per case	Low (code); high (user study)	Defensible reasons; regulator fit	Faithfulness checks; small analyst study on time and accuracy	Application grounded explanation study, if analysts are available
5. Multi layer risk assessment	Session and message cues target coerced payments where transactions look normal; untested, not refuted	RBA, JPMorgan journeys, bolewara text, BAF	Linked message, session and payment records; shared state store	High	Covers digital arrest and investment scams (85% of 2025 losses per I4C reporting)	Needs linked data; until then, per layer evaluation only	Only with partner data
6. Real time payment protection	Mandatory UK APP reimbursement shifted liability to PSPs, with 88% reimbursed in year one (PSR); India uses DoT FRI for payee risk (MediaNama)	Simulated UPI generator; FinBench for graph store load	Sharded state, Lua scripts, latency budget, FRI style payee input	High	Credible deployment story	p99 latency at target TPS; decision parity offline vs streaming	Systems paper on causal feature state at scale

The **realistic extensions**, achievable by one student in a semester on public data, are directions 1, 2, 3 and the engineering half of 4. Each produces a clean, reproducible result, and together they support a workshop paper titled along the lines of “Causal, serving parity evaluation of mule detection on AMLworld: what actually adds value”, with the stage ablations, the threshold transfer bug, the random versus chronological gap and cross generator transfer (IBM to SAML-D to AMLgentex) as contributions. The **ambitious long term ideas** are direction 5 (multi domain fusion on linked message, session and payment data), direction 6 at UPI scale, an analyst user study for direction 4, and federated or consortium learning across banks. All of them depend on data or partners the project does not have, which is precisely why they belong in the jury conversation rather than the next sprint.

Phase 7: twenty jury questions that trade numbers for access

Each question opens with a measured fact so the jury sees the team has done the work, then asks for something only an expert or institution can supply.

#	Question	Technical problem	Why we need the jury	Outcome we hope for
1	Our models drop from PR-AUC 0.716 to 0.336 when moved between two synthetic generators. Would your institution share a pseudonymised, time stamped extract of inbound UPI credits for three to six months, with mule labels from freeze orders, under NDA on your premises?	All evidence is synthetic; generalisation to real data is unknown	Only banks hold labelled receiving side data	A data access path or named contact for an on premise pilot
2	RBIH built MuleHunter.AI around 19 mule behaviours. Could RBIH share those behaviours at definition level, or run our 31 features blind inside its sandbox?	No public mule labels exist; we derive them from laundering paths	RBIH holds the only national mule model	A benchmark comparison or sandbox entry
3	Is there any route today, through NPCI, IDRBT or the FREE-AI sandbox proposal, for a student team to get synthetic UPI data calibrated to real graph statistics, including VPA and device fields?	Our schema has no UPI fields; a simulator needs real calibration targets	These bodies set the data access rules	Calibration statistics or sandbox access
4	In your bank, what exactly becomes the ground truth “mule” label (NCRP freeze request, internal STR, analyst confirmation), and how many days after the credit does it arrive?	We assume instant, perfect labels; production labels are late and noisy	Only practitioners know label provenance and delay	A label delay value to simulate, and a label definition to adopt
5	Mule chains cross banks and the payer PSP and payee bank each see half. Where should a model like ours sit: payer side, payee side, or switch level?	Partial observability; AMLgentex shows local views lose signal	Architecture depends on who can act on the score	A deployment position that defines our feature set

#	Question	Technical problem	Why we need the jury	Outcome we hope for
6	Our verify band flags 1.5% of payments at 10% precision. What daily alert volume per million UPI transactions can a fraud operations team absorb, and what step up rate do customers tolerate?	No alert budget or capacity constraint in our evaluation	Capacity is an operational fact, not a modelling choice	A precision at k target to optimise
7	We must choose a block threshold: 0.823 gives precision 0.64 at recall 0.65, while 0.9345 gives 0.94 at 0.55. Given the guidance against blocking on one weak signal, which would a bank accept, and should any single model ever trigger a hold?	Open block threshold decision	Risk appetite belongs to the bank	A threshold policy and a two domain rule we can cite
8	What cost ratio do you use between a missed mule credit and a false hold on a genuine customer, and should it scale with amount?	No cost sensitive evaluation exists	Costs include reputation and complaints only banks can price	A cost matrix for threshold selection
9	Our chronological test covers 18 simulated days. What minimum walk forward horizon and how many retraining cycles does your model risk team require before approval?	Short horizon cannot show drift	Model validation standards are internal	A validation protocol to adopt on AMLworld Large
10	Isolation Forest recovered 0 of four held out typologies while supervised XGBoost kept 0.80 to 0.85. How do your teams surface a new typology before labels exist?	Unseen pattern detection	Practitioners see emerging scams first	Typology intelligence or a feedback channel
11	Published GNNs on AMLworld report F1 up to 76, but our strictly causal GNN reaches PR-AUC 0.516. In production, is any same window future context permitted, or is strict past only scoring the norm?	Benchmark optimism vs deployable protocol	Defines whether graph benchmark results are relevant	Confirmation of the causal protocol as the standard
12	Would sharing HMAC tokens of confirmed mule VPAs across banks, as our signed ledger does, be acceptable under DPDP and RBI rules, or would it need federated learning?	Cross bank ring detection	Legal interpretation is outside our competence	A privacy design that a compliance team would accept
13	Can a third party consume DoT’s Financial Fraud Risk Indicator or NCRP flagged numbers as a payee risk feature, and at what latency?	We lack payee risk inputs	Access is controlled by DoT and banks	Integration feasibility for an FRI style input

#	Question	Technical problem	Why we need the jury	Outcome we hope for
14	What end to end latency budget does a payer PSP have for a risk call inside a UPI transaction, and at what peak TPS? Our API is p95 71.8 ms at 220 rps; our distributed path is p95 795 ms	No published UPI latency budget found	Infrastructure experts know the real constraint	A latency and throughput target
15	FREE-AI asks for systems “understandable by design”. Do investigators want SHAP reason codes, rule evidence, or graph paths, and does our explaining only the XGBoost half disqualify us?	Partial explanations	Regulators and investigators define adequacy	An explanation format specification
16	Would any bank run RuseTect in shadow mode for four weeks on mirrored traffic, with no customer impact, so we can measure precision against your existing alerts?	No measurement under real conditions	Only a bank can provide mirrored traffic	A shadow pilot
17	In real UPI switch records, which fields are routinely missing or unreliable (device ID, location, SIM age), and how should a model behave when they are?	We impute missing format as ACH, which shifts risk	Data quality patterns are internal knowledge	A missing field policy and test cases
18	Can banking apps lawfully collect screen sharing, paste and active call signals to stop digital arrest scams, and does any labelled dataset of coerced sessions exist?	Session layer has no benchmark	Legal and data questions only practitioners can answer	Consent design and possible data
19	RBI proposes testing AI fraud detection for accuracy and bias. Which attributes and parity metrics would an auditor check for mule flagging? We report age FPR parity 0.93 on BAF	Fairness audit scope unclear	Regulators define the test	An audit checklist
20	Would a faculty member jointly supervise a paper on causal, serving parity evaluation of mule detection with our negative results, and is ICAIF or a KDD applied data science workshop the right venue?	Publication needs mentorship and venue fit	Academic jurors know venues and reviewers	A supervisor and a target venue

If the interaction allows only five questions, ask **1, 4, 6, 14 and 16**, in that order. Question 1 attacks the single largest weakness (all data synthetic) and is the only one that could change the project’s trajectory. Question 4 fixes the definition of the target itself; without a real label definition and delay, every metric is

measured against the wrong thing. Question 6 converts the false positive debate into an operational target that the team can optimise and report. Question 14 sets the engineering bar for the whole real time story with one number. Question 16 is the concrete, low risk ask that turns interest into a pilot. Questions 2 and 7 are the best substitutes if the panel includes RBIH or a bank risk officer respectively.

Phase 8: a four level roadmap from threshold fix to research platform

The brief is cut off at “Research” in the field list; it is read here as “research basis”, and the requested evaluation method, effort and expected benefit are added. Effort is an estimate in person weeks for one student familiar with the codebase. File names refer to the existing repo.

Level 1: strengthen the existing system

Item	Technical objective	Required dataset	Required code changes	Research basis	Evaluation method	Effort	Expected benefit
1.1 Calibrate and fix the block band	Calibrate XGBoost and LightGBM separately, average calibrated scores, refit all bands on validation	HI and LI validation weeks	Calibrators in `ml/retrain_extended.py`; new thresholds in `cascade_config.json`; engine applies calibration	Niculescu-Mizil and Caruana; Dal Pozzolo 2015	Brier, ECE, reliability plot on test; block precision and recall	1	Removes the open bug; block precision near 0.94 if the retuned point holds
1.2 Dataset expansion	Add AMLworld HI-Medium and SAML-D; cross generator train and test	HI-Medium, SAML-D	SAML-D column mapping loader; config per dataset	Altman et al.; Oztas et al.	PR-AUC matrix train on X, test on Y	2	Honest generalisation estimate beyond one generator
1.3 Improved validation	Bootstrap intervals, five seeds, random vs chronological comparison, label delay simulation	HI-Small, HI-Medium	Evaluation harness in ml/ with resampling and a shuffled split mode	Maganti; Dal Pozzolo 2018	Confidence intervals on every headline; size of random split optimism	1	Every claim becomes defensible; a strong jury slide
1.4 Feature engineering	Add past only GFP style fan in, fan out, scatter gather, cycle counts and windowed vertex stats	HI, LI, Medium	New incremental counters in streaming state; parity test extended	GFP; FlowScope	Ablation vs 31 features under the existing adoption rule; per typology recall	2 to 3	Likely gain on BIPARTITE and multi hop rings

Item	Technical objective	Required dataset	Required code changes	Research basis	Evaluation method	Effort	Expected benefit
1.5 Better benchmarking	Replace degenerate rules baseline with tuned expert rules; report precision at k per day, recall at fixed FPR, amount weighted and cost based metrics	HI, LI	Rules tuned on validation with sane bounds; metrics module	Fraud handbook; Bahnsen et al.	Same split, side by side table	1	Fair comparison; metrics a bank recognises
1.6 Error analysis	Review false negatives and false positives by typology, amount, account age; compute ring level precision and share of ring payments caught	HI, LI test	Analysis notebook reading results JSON	Fraud handbook	Error taxonomy; ring precision	1	Targets the next feature work
1.7 Hygiene and claims	Fix stale docs, replace ACH default with explicit missing indicator, compute SHAP over full score, rewrite the paper	none new	`server.py`, `rusetect_engine.py`, docs, paper	Lundberg et al.	Missing field test; reason faithfulness check	1	Removes the jury's easiest attacks

Level 2: advanced fraud intelligence

Item	Technical objective	Required dataset	Required code changes	Research basis	Evaluation method	Effort	Expected benefit
2.1 Graph based detection	Causal TGN or FraudGT research arm compared with GFP plus XGBoost	AMLworld S/M; Elliptic++ for real labels	Separate offline module with per decision past only neighbour sampling	FraudGT; TGN; Maganti	Identical causal split; adopt only if validation gain clears the rule	3 to 4	A rigorous answer to "why not a GNN"
2.2 Temporal transaction modelling	Longer windows and a fused sequence embedding feature	AMLworld Large; TabFormer for pretraining	Window config; embedding export to features	nuFormer	Late fusion ablation with intervals	2	Small expected gain; tests a popular claim
2.3 Mule account detection	Account level labels and model (fan in then rapid fan out) alongside payment model	AMLworld, SAML-D, AMLgentex	Account aggregation job; account score fed to payment model and ring linker	CFDB; MuleHunter context	Account PR-AUC; time to first detection after first mule credit	2 to 3	Matches how banks act (freeze accounts, not payments)

Item	Technical objective	Required dataset	Required code changes	Research basis	Evaluation method	Effort	Expected benefit
2.4 Behavioural anomaly detection	Login and device risk prior for account takeover; autoencoder kept as novelty monitor only	RBA, JPMorgan journeys, BrainRun	Session layer model trained on RBA; monitor dashboard	RBA dataset; ADBench	ATO ROC and FPR on RBA; novelty alert precision	2	First measured session layer
2.5 Explainable predictions	Reasons over full score plus graph path and rule evidence per case	Existing results	SHAP for both boosters; case view showing ring paths	Jesus et al.; Weerts et al.	Faithfulness (deletion test); optional small analyst study	1 to 2	Regulator ready explanations

Level 3: realistic banking environment

Item	Technical objective	Required dataset	Required code changes	Research basis	Evaluation method	Effort	Expected benefit
3.1 Streaming transaction processing	Shared state store so message, session and payment meet on one instance; Lua scripts and pipelining for Redis	Replay of HI-Medium	`service/scale/`; replace in memory state on the hosted demo	GFP streaming design	Throughput and p99 at target TPS; decision parity vs offline	3	Fixes the “fusion silently does nothing” defect
3.2 Real time inference	Meet a stated latency budget with retrained models	Replay	Profiling; rerun online parity on current models	Fraud handbook	p50, p95, p99 per payment; load test at multiple worker counts	1 to 2	Replaces the stale 0.33 ms claim
3.3 Risk based alerting	Capacity aware bands: review queue with daily k, learning to defer	HI, LI	Band logic in engine; queue in review desk	Alves et al.	Precision at k per day; missed laundering per analyst hour	2	Directly answers alert fatigue
3.4 Investigation workflows	Case packages with evidence, two analyst approval, customer hearing and appeal before freeze	Ring cases from results	`review.py`, `security.py`, `str_report.py`	RBI Master Directions 2024	Case resolution time in a scripted walkthrough	2	Aligns with natural justice requirement
3.5 Model monitoring	Model card, inventory entry, incident log, score distribution and alert rate monitors	Replay	Monitoring module; documentation	FREE-AI via Mondaq	Alarms fire on injected faults	1 to 2	Regulator facing artefacts
3.6 Drift detection	Detect and recover from typology change	AMLgentex step 56 switch; AMLworld Large; BAF months	Drift statistics on features and scores; retraining trigger	Dal Pozzolo 2018; Lebichot et al.	Detection delay after switch; recall recovery after retrain	2	Evidence for “what happens when behaviour changes”

Level 4: research grade platform

Item	Technical objective	Required dataset	Required code changes	Research basis	Evaluation method	Effort	Expected benefit
4.1 Reproducible experiments	One command per result, pinned versions, seeds, data cards, results with hashes	All in use	Experiment runner and config files in ml/	Fraud handbook reproducibility focus	Clean machine rerun matches reported numbers	2	Reviewers can verify every claim
4.2 Multiple benchmark datasets	Standard suite: AMLworld S/M/L, SAML-D, AMLgentex, Elliptic++, DGraph-Fin	As listed	Loaders per dataset to the common schema	Altman et al.; Östman et al.; Elliptic++	Results table across suite	3 to 4	Generalisation claims become testable
4.3 Standardised evaluation	Fixed protocol: causal splits, label delay, bootstrap, alert budget, cost, latency	Suite above	Shared evaluation library	Saito and Rehmsmeier; Maganti	Protocol documented and applied to all models	2	The core of a methodology paper
4.4 Generalisation testing	Cross generator and synthetic to real label transfer	IBM to SAML-D to AMLgentex; Elliptic++, Libra	Transfer experiments	Maganti	Drop in PR-AUC per transfer	2	Quantifies the synthetic data gap
4.5 Privacy preserving architecture	Measure ring recall lost when graph features are bank local; prototype federated training	AMLworld partitioned by bank; AMLgentex 12 institutions	Partitioning; federated averaging prototype	PETs Prize; AMLgentex	Ring recall local vs pooled vs federated	3 to 4	Evidence for cross bank sharing value
4.6 Deployment and scalability testing	Graph store and ring query load, end to end stress at UPI like rates	LDBC FinBench; replayed AMLworld Large	Containerised deployment tested (docker compose currently untested)	FinBench	Sustained TPS and p99; ring query latency	3	Credible capacity numbers

Level 1 is roughly ten person weeks and should be finished before any new modelling, because it converts every existing number into a defensible one. Level 2 depends on Level 1’s evaluation harness. Level 3 is where partner input (questions 6, 14, 16) changes the targets. Level 4 is what turns the work into a publishable benchmark.

Conclusion

The most important change in understanding is that RuseTect is not a multistage detector that needs more stages; it is a well engineered single model whose features carry the signal, wrapped in layers that are designed for data the project does not yet have. That reframing is a strength in front of an expert jury: the team can show that it measured each layer, kept what helped and dropped what did not, which is rarer than a high headline number. The weakness a juror will

find in minutes is not the model but the gap between the evidence and the claims, so the submission materials must be brought into line with the repo before the finale.

The practical implication is that the next large gain will not come from architecture. It will come from three things only outsiders can provide or force: real or realistically calibrated data with real label definitions and delays, an operational target (alert budget, cost ratio, latency budget) that defines what “good” means, and a second generator or real label set that tests whether 0.711 survives outside IBM’s simulator. The 20 questions and the Level 1 roadmap are designed so that every conversation with the jury either produces one of those three things or produces a result that makes the project more publishable without them.